

Sawtooth WAVE GENERATOR

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A sawtooth waveform is a type of linear, non-sinusoidal waveform with a triangular shape in which the rise time and fall time are different. A pure triangular waveform is also linear and non-sinusoidal, but it has equal rise and fall times. The sawtooth waveform can also be called an asymmetric triangular wave.

Sawtooth waveforms can be generated by integrating square waves with unequal rise and fall times (asymmetrical square waves). Triangular waveforms and sawtooth waveforms are used in applications like thyristor (SCR, triac, etc) switching, sampling, tone/frequency generation, modulation, etc. A sawtooth wave generator using the NE555 timer is presented here. The prototype tested in EFY lab is shown in Fig. 1.

Fig. 2 shows the circuit diagram of the sawtooth wave generator. The heart of the circuit is the NE555 timer IC, which is used for generating the sawtooth waves. This is a very simple and inexpensive circuit for generating sawtooth waveforms. Normally, UJT-based relaxation oscillators are used, but they are not freely available and are also expensive.

This circuit uses just the NE555 timer IC and a few passive components. We know that the NE555 timer IC is generally used to generate a square wave signal. But with slight modifications at the output, sawtooth waveform generation is possible.

The IC NE555 is wired as an astable multivibrator with unequal on and off times. The resistors R1 and R2 and capacitor C3 set the on and off time periods. The values of these components are such that the off time is less than the on time.

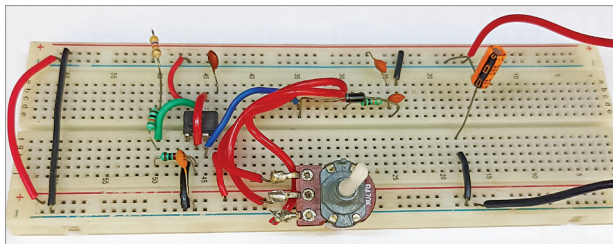


Fig. 1: EFY lab's prototype on breadboard

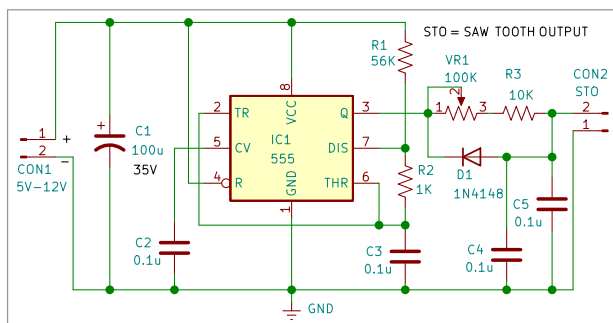


Fig. 2: Circuit diagram

When the power supply is switched on, capacitor C3 starts charging through resistors R1 and R2. When the voltage across C3 rises above $2/3V_{CC}$, the upper comparator inside the NE555 swings to positive saturation, triggering the internal flip-flop. This makes the output pin 3 of the timer go high. Now capacitor C3 starts to discharge through resistor R2 into pin 7 of the IC. When the voltage across capaci-

tor C3 becomes less than $1/3V_{CC}$, the lower comparator inside the IC switches to positive saturation, triggering the internal flip-flop again. As a result, the output of the timer at its pin 3 goes low. This action is repeated, resulting in a square wave at pin 3 of the NE555. The charging time (on time) is given by the equation $T_1 = 0.69(R_1 + R_2)C_3$, and the discharging time (off time) is given by the equation

$$T_2 = 0.69R_2C_3.$$

This square wave output at pin 3 of the NE555 is further processed through potentiometer VR1, resistor R3, signal diode 1N4148 (D1), and capacitors (C4, C5) to obtain a sawtooth signal across CON2. Potentiometer VR1 can be adjusted to get the maximum output of the sawtooth wave.

Assemble the circuit on a breadboard or a general-purpose PCB and enclose it in a small cabinet. Fix CON1 at the rear side of the box and connect a 12V battery or adaptor for the power supply. Connect CON2 at the front of the box for using the sawtooth output. Now the sawtooth generator is ready to use. **EFY**

PARTS LIST

Semiconductors:

- IC1 - 555/NE555 timer IC
- D1 - 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 56-kilo-ohm
- R2 - 1-kilo-ohm
- R3 - 10-kilo-ohm
- VR1 - 100-kilo-ohm potentiometer

Capacitors:

- C1 - 100 μ F, 35V electrolytic
- C2-C5 - 0.1 μ F ceramic disk

Miscellaneous:

- CON1, CON2 - 2-pin connector

Pradeep G. is an electronics designer based in Cochin, Kerala. His areas of interest are digital, opto electronics, wireless communications and remote control projects and all types of sensor based projects.

